HOLOMORPHIC $\mathcal{N}=1$ VERTEX OPERATOR SUPERALGEBRAS OF MINIMAL INDEX

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ABSTRACT. Gaiotto and Johnson-Freyd asked whether for every k there is a holomorphic $\mathcal{N}=1$ vertex operator superalgebra of central charge $12k$ whose supersymmetric index is $24/\gcd(k,24)$, the smallest value compatible with topological modular forms. They answered the question for $k \leq 5$ by orbifolding the lattice superalgebra of a self-dual ternary code of index 24 by a cyclic group of code automorphisms. Using the codes of a companion paper we answer it for every k, within their orbifold framework. When $8 \nmid k$ their cyclic construction applies. When $8 \mid k$ no cyclic group can work, and we orbifold instead by the quaternion group Q_8 ($k \equiv 8, 16 \pmod{24}$) or by $Q_8 \times \mathbb{Z}/3$ ($k \equiv 0 \pmod{24}$). The main new input is the 't Hooft anomaly of a quaternion group of code automorphisms: it vanishes if and only if $8 \mid k$. Its restrictions to cyclic subgroups vanish already for $4 \mid k$; one class of restricted supercohomology is invisible to all of them, and we detect it through the (-1)-twisted sector, by showing that the character of a would-be orbifold fails to be S-invariant when $k \equiv 4 \pmod{8}$. We also show that the orbifold has index $\mathrm{Index}(C)/|G|$ for every group G of code automorphisms whose abelian subgroups are cyclic, whose only involution is -1, and whose elements of odd order form a central subgroup of order at most 3 that fixes no coordinate and as many even as odd maximal codewords. For $k = 24$ this gives a holomorphic $\mathcal{N}=1$ vertex operator superalgebra of central charge 288 and index 1.

1. INTRODUCTION

A self-dual ternary code $C \subseteq \mathbb{F}_3^n$ determines the odd unimodular lattice $\Lambda(C) = \{a/\sqrt{3} : a \in \mathbb{Z}^n,\ a \bmod 3 \in C\}$ and hence a holomorphic vertex operator superalgebra $V_{\Lambda(C)}$ of central charge n, which Gaiotto and Johnson-Freyd equip with an $\mathcal{N}=1$ structure [14, §2]. Its supersymmetric (Ramond–Ramond) index is the index of the code [14, Prop. 2.4],

$$\mathrm{Index}(C) = \#\{\text{even maximal codewords}\} - \#\{\text{odd maximal codewords}\},$$

where a codeword is maximal if none of its entries is 0, and even or odd according to the parity of the number of its entries equal to 1. Under the Stolz–Teichner conjecture a holomorphic $\mathcal{N}=1$ vertex operator superalgebra of central charge $12k$ defines a class in $\pi_{24k}\mathrm{TMF}$ whose image in MF_{12k} is its index times Δ^k [14, §5], and the smallest multiple of Δ^k in the image of $\pi_{24k}\mathrm{TMF} \to \mathrm{MF}_{12k}$ is $\frac{24}{\gcd(k,24)}\Delta^k$ [15, Prop. 4.6]. Gaiotto and Johnson-Freyd ask whether these values are attained:

Main Question ([14, §0, §5])**.** *For each k, does there exist a holomorphic $\mathcal{N}=1$ VOA with central charge $c = 12k$ and supersymmetric index $\frac{24}{\gcd(k,24)}$?*

The index is defined up to sign, the parity of the Ramond sector being a convention [14, Rem. 5.3]; "index m" means index $\pm m$ throughout. Gaiotto and Johnson-Freyd answered the question for $k \leq 5$ with the following theorem, found their codes by computer search, and asked for a systematic construction [14, Question 0.1, Thm. 5.6, Ex. 5.7].

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> **Theorem** ([14, Thm. 5.6]). Let C be self-dual of length $12k$ with $\mathrm{Index}(C) = 24$, put $m = \gcd(24, k)$, and let $g \in \mathrm{Aut}(C)$ have order m. Suppose that (1) if m is even, $g^{m/2} = -1$, and (2) if $3 \mid m$, $g^{m/3}$ fixes as many even as odd maximal codewords and permutes the coordinates in 3-cycles only. Then $V_{\Lambda(C)}/\!\!/\langle g\rangle$ has index $24/m$.

In a companion paper [16] we constructed explicit self-dual codes of index 24 in every length $12k$, with the automorphisms that this theorem and the present paper need, and we showed that for $8 \mid k$ no cyclic group can work: an automorphism g with $g^4 = -1$ forces $48 \mid \mathrm{Index}(C)$ [16, Thm. D]. Here we supply the vertex-algebra side.

Theorem 1.1 (Main Theorem)**.** *Assume Hypothesis ORB (§2.6) and the cited inputs (A1)–(A10) of §2. For every $k \geq 1$ there is a holomorphic $\mathcal{N} = 1$ vertex operator superalgebra of central charge $12k$ and index $24/\gcd(k, 24)$, namely $V_{\Lambda(C)}/\!\!/G$ for a code C and a group $G \subseteq \mathrm{Aut}(C)$ as in Table 1.*

$\gcd(k, 24)$	k mod 24	index	G	code	seed of the code
1	$\pm1, \pm5, \pm7, \pm11$	24	1	[16, Thm. B]	Golay
2	$\pm2, \pm10$	12	$\langle -1\rangle$	[16, Thm. B]	Golay
3	$\pm3, \pm9$	8	$\langle h\rangle \cong \mathbb{Z}/3$	[16, Thm. C(a)]	E_{36} of [14]
4	±4	6	$\langle J\rangle \cong \mathbb{Z}/4$	[16, Thm. C(b)]	Golay
6	±6	4	$\langle -h\rangle \cong \mathbb{Z}/6$	[16, Thm. C(a)]	E_{36} of [14]
8	±8	3	$Q_8 = \langle J, K\rangle$	[16, Thm. C(b)]	Golay
12	12	2	$\langle Jh\rangle \cong \mathbb{Z}/12$	[16, Thm. C(c)]	E_{72} of [16, App. A.2]
24	0	1	$Q_8 \times \langle h\rangle$	[16, Thm. C(c)]	E_{72} of [16, App. A.2]

TABLE 1. The orbifolds of Theorem 1.1. Here $J^2 = K^2 = -1$, $KJ = -JK$, and h has order 3, no fixed coordinate, a vanishing twined count $T_h(C) = 0$, and commutes with J and K.

For $8 \nmid k$ this is [14, Thm. 5.6] applied to the codes of [16], as noted there. Sections 3 and 6 reprove both halves of that theorem, the vanishing of the anomaly and the value of the index, from the same inputs and in a form that also covers the non-cyclic groups. For $8 \mid k$ the new ingredients are the following three theorems. All automorphisms of codes act on $V_{\Lambda(C)}$ through the lift of [14, Thm. 3.1].

Theorem 1.2. *Assume (A1)–(A10) and Hypothesis ORB. Let C be a self-dual ternary code of length $12k$, and let $J, K \in \mathrm{Aut}(C)$ satisfy $J^2 = K^2 = -1$ and $KJ = -JK$. The action of $Q_8 = \langle J, K\rangle$ on $V_{\Lambda(C)}$ is non-anomalous if and only if $8 \mid k$.*

Theorem 1.3. *If moreover $h \in \mathrm{Aut}(C)$ has order 3, no fixed coordinate, and commutes with J and K, then $Q_8 \times \langle h\rangle$ acts non-anomalously on $V_{\Lambda(C)}$ if and only if $24 \mid k$.*

Theorem 1.4. *Let C be self-dual of length $12k$ and $G \subseteq \mathrm{Aut}(C)$ a subgroup such that*

(i) *every abelian subgroup of G is cyclic;*
(ii) *if $|G|$ is even, then $-1 \in G$ is the only involution of G;*
(iii) *the elements of odd order form a central subgroup of order 1 or 3, and in the second case it is generated by an h without fixed coordinate such that $T_h(C) = 0$.*

If G acts non-anomalously on $V_{\Lambda(C)}$, then, under Hypothesis ORB, every orbifold $W = V_{\Lambda(C)}/\!\!/G$ is a holomorphic $\mathcal{N} = 1$ vertex operator superalgebra with $\mathrm{Index}(W) = \pm\,\mathrm{Index}(C)/|G|$.

Here $T_h(C)=\sum_{x\in C,\,hx=x} f(x)$ is the signed count of h-fixed maximal codewords (§2.1); it vanishes iff h fixes as many even as odd maximal codewords. All groups of Table 1 satisfy (i)–(iii).

The idea for Q_8. The anomaly of a Q_8-action lies in the restricted supercohomology group $\mathrm{rSH}^3(Q_8)$, of order 32. The Frame-shape computation of [14, Ex. 4.4] gives its restrictions to the cyclic subgroups, which vanish iff $4\mid k$. A count (Proposition 4.2) shows that exactly one nonzero class κ restricts to zero on every cyclic subgroup. Since every abelian subgroup of Q_8 is cyclic, no twisted–twined partition function of a commuting pair can see κ. We detect it through the (-1)-twisted sector. As a projective Q_8-module this sector does not depend on C: it is the spinor module of the Clifford algebra of the frame's zero modes (Lemma 4.4). So the character that a Q_8-orbifold would have can be written down, and three eta identities show that it is not S-invariant when $k\equiv 4 \pmod 8$ (Proposition 4.6). For $8\mid k$, additivity of anomalies under direct sums of codes finishes the proof. For the record, the anomaly is $3k\cdot a$, where a is the anomaly of eight Majorana fermions in the real regular representation of Q_8, a class of order 8 that takes the value $\frac{5}{8}$ on S^3/Q_8 (Remark 4.7).

What the results rest on. The fermionic case of the orbifold theorem [14, Thm. 4.2] is only sketched in [14]; for bosonic vertex operator algebras it is a theorem [17, 8, 20]. Hypothesis ORB (§2.6) states what we use from it: existence of the orbifold for a non-anomalous group, and the decomposition of its NS and Ramond sectors into invariants of twisted sectors, with genuine actions of the centralisers. We also use the description of the Majorana layer and the multiplier principle for cyclic groups from [14, §4], which are stated there without proof, and standard theorems on lattice vertex superalgebras and their twisted modules [2, 19, 10, 9]. Gaiotto and Johnson-Freyd's own answers for $k\le 5$ rest on the same inputs. On the code side we use [16]; for $12\mid k$ one numerical input enters there, the value 24 of a block-weighted sum over a code of length 72, which was computed by three independent programs.

Two remarks. Holomorphic $\mathcal{N}=1$ vertex operator superalgebras are closed under tensor products, and for central charges divisible by 12 the index is multiplicative. So the theory of central charge 288 and index 1 in Theorem 1.1 reduces the Main Question to $k\le 24$ (Remark 7.1). Under the Stolz–Teichner conjecture that theory represents a unit of $\pi_{576}\mathrm{TMF}$ (Remark 7.2).

Organisation. §2 collects the setting and the cited inputs. §3 treats cyclic groups, §4 proves Theorem 1.2, §5 Theorem 1.3, and §6 Theorem 1.4. §7 proves the Main Theorem and adds remarks. The appendix lists the computations that check the explicit formulas.

2. Setting

2.1. **Codes.** We follow [16, §2]. Let $\varphi\colon \mathbb{F}_3\to\{0,\pm1\}$ be $\varphi(0)=0$, $\varphi(1)=-1$, $\varphi(2)=1$, let $f(x)=\prod_i\varphi(x_i)$, and $F(X)=\sum_{x\in X}f(x)$ for $X\subseteq\mathbb{F}_3^n$. Then $\mathrm{Index}(C)=F(C)$. The group $\mathrm{Mon}(n)=2^n{:}S_n$ acts on $\mathbb{F}_3^n$ and on $\mathbb{R}^n$ by $ge_i=s_ie_{p(i)}$ for $g=(p,s)$, and $\mathrm{Aut}(C)=\{g: gC=C\}$ contains -1. Put $N_-(g)=\#\{i: s_i=-1\}$ and $\varepsilon(g)=\prod_i s_i$; then ε is a character and $f(gx)=\varepsilon(g)f(x)$.

A cycle of p of length ℓ whose signs have product σ is a *signed cycle*; it contributes the factor $1-x^\ell$ ($\sigma=1$) or $1+x^\ell=(1-x^{2\ell})/(1-x^\ell)$ ($\sigma=-1$) to $\det(1-xg)=\prod_k(1-x^k)^{d_k}$. The formal product $\prod_k k^{d_k}$ is the *Frame shape* of g, and we put $\eta_g(\tau)=\prod_k\eta(k\tau)^{d_k}$ and

$$\rho_g=\tfrac14\sum_j\theta_j(1-\theta_j)=\tfrac{1}{24}\sum_k d_k\big(k-\tfrac1k\big),$$

where $e^{2\pi i\theta_j}$, $\theta_j \in [0,1)$, are the eigenvalues of g on $\mathbb{R}^n$. The vector space $\mathbb{R}^n$ has no nonzero g-fixed vector iff every signed cycle has $\sigma = -1$, iff $\sum_k d_k = 0$. We say that g has *no fixed coordinate* if p has no fixed point; this is weaker (an element of order 3 without fixed coordinate fixes the vectors that are constant on its cycles). On a signed ℓ-cycle, g^j is diagonal iff $\ell \mid j$, and then it is $\sigma^{j/\ell}$. Hence

$$g^j = -1 \iff \text{every signed cycle has length } \ell \mid j \text{ with } j/\ell \text{ odd and } \sigma = -1. \tag{1}$$

For example $J^2 = -1$ means that J consists of $n/2$ signed 2-cycles with $\sigma = -1$, so $N_-(J) = n/2$, J has no fixed vector, and its Frame shape is $2^{-n/2}4^{n/2}$.

2.2. **The vertex superalgebra of a code.** Let C be self-dual of length $n = 12k$, $\Lambda = \Lambda(C)$ and $V = V_\Lambda$. The *frame* is the subalgebra $V_{\sqrt{3}\mathbb{Z}^n} = V_{\sqrt{3}\mathbb{Z}}^{\otimes n} \subseteq V$. The vector $\chi = \sqrt{3}(1,\dots,1)$ is characteristic: $\langle\chi,\lambda\rangle \equiv |\lambda|^2 \pmod 2$ for $\lambda \in \Lambda$. Let σ be the parity automorphism of V.

Proposition 2.1 ([14, §2, Prop. 2.4]; [2, 19, 10])**.** (1) *V is a strongly rational holomorphic vertex operator superalgebra of central charge n.*

(2) *$\tau = \frac{1}{\sqrt{3}}\sum_i\big(e^{\sqrt{3}e_i} + e^{-\sqrt{3}e_i}\big)$, for a suitable cocycle, is an $\mathcal{N} = 1$ supercurrent: $\tau_{(0)}\tau = 2\omega$, $\tau_{(1)}\tau = 0$, $\tau_{(2)}\tau = \frac{2n}{3}\mathbf{1}$.*

(3) *The Ramond sector V_R, the unique σ-stable irreducible σ-twisted module, is $V_{\Lambda+\chi/2}$, with parity $(-1)^{\langle\chi,\lambda\rangle}$ on $M(1)\otimes e^{\lambda+\chi/2}$.*

(4) *The states of V_R of weight $c/24 = n/24$ are the vectors e^{ν_w}, one for each maximal codeword w, with $\nu_w = \iota(w)/(2\sqrt{3})$ and $\iota\colon \mathbb{F}_3 \to \{0,1,-1\}$; the parity of e^{ν_w} is $f(w)$.*

(5) *On every g-twisted V-module with $g\tau = -\tau$ the modes $G_r = \tau_{r+1/2}$, $r \in \mathbb{Z}$, satisfy $G_0^2 = L_0 - c/24$. In particular $\operatorname{str}_{V_R} q^{L_0 - c/24} = \operatorname{Index}(C)$.*

Comments. (1) is [14, §2] with [2, Thm. 4.1] for odd lattices; (3) is [19, Prop. 5.4] applied to $\chi/2$, with uniqueness from [10, Thm. 7.7]. (4): a Ramond momentum is $\nu = (2a + 3\cdot\mathbf{1})/(2\sqrt{3})$ with $a \in \mathbb{Z}^n$, $a \bmod 3 \in C$, so $|\nu|^2/2 = \sum_i(2a_i+3)^2/24 \ge n/24$, with equality iff every $a_i \in \{-1,-2\}$. Then $w = -(a \bmod 3)$ is maximal, $\nu = \nu_w$, and the parity $(-1)^{\sum a_i}$ is $(-1)^{\#\{i:w_i=1\}} = f(w)$. (5) is the twisted commutator formula [19, (M4)] with (2); if G_0 is odd, invertible on each weight space of weight $\ne c/24$, and commutes with an operator A, then A has supertrace 0 there. So supertraces of operators commuting with G_0 only see weight $c/24$, and (4) gives the last claim. □

2.3. **Automorphisms.** By [14, Thm. 3.1] the extension $\widehat{\Lambda}[2].\operatorname{Aut}(C)$ of $\operatorname{Aut}(C)$ inside $\operatorname{Aut}(V)$ splits. We write $s\colon \operatorname{Aut}(C) \to \operatorname{Aut}(V)$ for the splitting that restricts to the standard action of $\operatorname{Mon}(n)$ on the frame, $e^{\pm\sqrt{3}e_i} \mapsto e^{\pm s_i\sqrt{3}e_{p(i)}}$. It is unique: the restriction $\operatorname{Hom}(\Lambda,\pm1) \to \operatorname{Hom}(\sqrt{3}\mathbb{Z}^n,\pm1)$ is bijective, since $\sqrt{3}\mathbb{Z}^n$ has odd index in Λ. Consequently $s_{C\oplus C'} = s_C \otimes s_{C'}$ on $V_{\Lambda(C\oplus C')} = V_{\Lambda(C)} \otimes V_{\Lambda(C')}$. Every $s(g)$ fixes τ, so $s(G)$ acts by even automorphisms, faithfully, with $s(G) \cap \langle\sigma\rangle = 1$; and $s(g)$ maps $M(1)\otimes e^\lambda$ to $M(1)\otimes e^{g\lambda}$ with a sign [2, Prop. 4.1]. We usually write g for $s(g)$.

Lemma 2.2. *For $g \in \operatorname{Aut}(C)$, $\langle\chi,(1-g)\chi\rangle = 6N_-(g)$, and $s(g)$ maps the summand of V_R with momentum $\lambda + \chi/2$ to the one with momentum $g(\lambda+\chi/2)$, multiplying the parity by $(-1)^{N_-(g)} = \varepsilon(g)$.*

Proof. $\langle\chi,\chi\rangle = 3n$ and $\langle\chi, g\chi\rangle = 3\sum_i s_i = 3(n - 2N_-(g))$. The relative parity is $\frac12\langle\chi,(1-g)\chi\rangle = 3N_-(g)$ modulo 2 [14, Ex. 4.4]. □

If $g \in O(\Lambda)$ has no fixed vector, any two lifts of g to $\widehat{\Lambda}.O(\Lambda)$ are conjugate by an element of the torus $\widehat{\Lambda} = \operatorname{Hom}(\Lambda, U(1))$ [14, Ex. 4.4]: if $t \in \widehat{\Lambda}$, then $t\hat{g}t^{-1} = \hat{g}\cdot(t\circ(g-1))$, and $t \mapsto t\circ(g-1)$

is onto because $g-1$ is injective on Λ and $U(1)$ is divisible. In particular every lift of such a g has the same anomaly as $s(g)$, and $\hat{g}\sigma$ is conjugate to $\hat{g}$, since $\sigma = e^{2\pi i(\chi/2)(0)}$ lies in the torus.

2.4. **Twisted modules.** For an automorphism g of finite order of a strongly rational holomorphic vertex operator superalgebra there is a unique σ-stable irreducible g-twisted module $V(g)$ [10, Thm. 7.7], [9, Prop. 3.4]. The *g-twisted Ramond sector* is $V_{R,g} := V(g\sigma)$. An *x-intertwiner* on a g-twisted module M, for x commuting with g, is an invertible A with $AY_M(v,z)A^{-1} = Y_M(xv,z)$; on $V(g)$ it exists and is unique up to a scalar when $V(g)$ is of $\mathbb{C}$-type, and then it is even or odd.

Proposition 2.3 ([2, Thm. 4.2, Rem. 4.4], [19])**.** *Let $g \in \mathrm{Aut}(C)$ have no fixed vector and $\varepsilon(g) = 1$. Then $V(g)$ is of $\mathbb{C}$-type, $V(g) \cong V(g\sigma)$ as graded spaces, and $V(g) = S[g] \otimes T_g$, where $S[g]$ is the g-twisted Fock space of the Heisenberg algebra and T_g is a space of ground states of dimension $|\det(1-g)|^{1/2} = \prod_k k^{d_k/2}$. The lowest weight is ρ_g, and $\operatorname{ch} V(g) = |\det(1-g)|^{1/2} \prod_k \eta(\tau/k)^{-d_k}$. Moreover $Z(1,g) := \operatorname{tr}_V g\, q^{L_0 - c/24} = 1/\eta_g(\tau)$.*

Comments. Only $\lambda = 0$ contributes to $Z(1,g)$, and the trace on the Fock space is $\prod_m \det(1 - gq^m)^{-1}$. The number of irreducible g-twisted modules is 1 or 2 according as $\frac{1}{2}\langle\chi, (1-g)\chi\rangle$ is even or odd [2, Thm. 4.2], i.e. according to $\varepsilon(g)$ (Lemma 2.2); the defect formula of [2] gives $\dim T_g$. The vacuum energy of $S[g]$ is ρ_g [12, Thm. 1.7.3], [18, 7]; equivalently, $\operatorname{ch} V(g)$ is the S-transform of $Z(1,g)$ [10, Thm. 7.1], whose leading exponent is $\rho_g - n/24$. Finally $V(g\sigma) \cong V(g)$ because $g\sigma$ is conjugate to g in the torus. □

2.5. **Anomalies.** We use the following statements from [14] and the literature it cites.

(A1) [14, Thm. 4.2]. For a finite group G acting faithfully by even automorphisms on a holomorphic vertex operator superalgebra V with V^G regular there is an anomaly $\alpha(V,G) \in \widetilde{\mathrm{SH}}^3(G)$, in reduced extended supercohomology. Orbifolds $V/\!\!/G$ exist iff $\alpha = 0$, and then correspond to trivialisations. For solvable G, regularity of V^G follows from [4] applied to the even part of V [14, §4].

(A2) The anomaly is natural: $\alpha(V,H) = \operatorname{res}^G_H \alpha(V,G)$ for $H \subseteq G$, and it is transported along isomorphisms of groups.

(A3) Conjugate actions have equal anomalies.

(A4) Anomalies add under tensor products: $\alpha(V \otimes V', \rho \otimes \rho') = \alpha(V,\rho) + \alpha(V',\rho')$ [13, §5].

(A5) There are exact sequences, natural in G,

$$\mathrm{H}^3(G;U(1)) \xrightarrow{\iota_1} \mathrm{rSH}^3(G) \xrightarrow{\mathrm{gw}} \mathrm{H}^2(G;\mathbb{Z}/2) \to \mathrm{H}^4(G;U(1)),$$
$$\mathrm{rSH}^3(G) \xrightarrow{\iota_2} \widetilde{\mathrm{SH}}^3(G) \xrightarrow{\mathrm{maj}} \mathrm{H}^1(G;\mathbb{Z}/2),$$

with ι_1 and ι_2 injective [14, §4], [26], [13, §5.4]. The maps maj and gw are the Majorana and Gu–Wen layers.

(A6) $\mathrm{maj}(\alpha)(g) = 0$ iff an operator on V_R implementing g is even [14, §4]. For the free fermions $\mathrm{Fer}(\rho)$ in an orthogonal representation ρ this gives $\mathrm{maj} = w_1(\rho) = \det \rho$.

(A7) (Multiplier principle [14, §4].) Let $G = \langle g\rangle \cong \mathbb{Z}/m$ act with vanishing Majorana layer, and put $L = m$ for m even, $L = 2m$ for m odd. Then $Z(1,g)/Z(1,1)$ is an eigenfunction of ST^LS^{-1}, and its eigenvalue $e_m(\alpha)$ depends only on $\alpha \in \mathrm{rSH}^3(\mathbb{Z}/m)$, through a homomorphism $e_m\colon \mathrm{rSH}^3(\mathbb{Z}/m) \to U(1)$.

For $G = \mathbb{Z}/m$ let $\nu_{m,r} \in \mathrm{rSH}^3(\mathbb{Z}/m)$ be the anomaly of the action on $\mathrm{Fer}(2) = V_{\mathbb{Z}}$ in which the generator rotates the two fermions through $2\pi r/m$, and $\nu_m = \nu_{m,1}$. Put $m' = 2m$ for m even and $m' = m$ for m odd.

Lemma 2.4. (1) $|\mathrm{rSH}^3(\mathbb{Z}/m)| = m'$, $|\mathrm{rSH}^3(Q_8)| = 32$, *and* $\mathrm{rSH}^3(\mathbb{Z}/m) = \widetilde{\mathrm{SH}}^3(\mathbb{Z}/m)$ *for* m *odd.*

(2) $e_m(\nu_{m,r}) = e^{2\pi i r^2/m'}$. *Hence* e_m *is injective,* $\mathrm{rSH}^3(\mathbb{Z}/m) = \langle \nu_m \rangle \cong \mathbb{Z}/m'$, *and* $\nu_{m,r} = r^2\nu_m$.

Proof. (1) For G cyclic or Q_8, $\mathrm{H}^3(G;U(1)) \cong \mathrm{H}^4(G;\mathbb{Z}) \cong \mathbb{Z}/|G|$ and $\mathrm{H}^4(G;U(1)) \cong \mathrm{H}^5(G;\mathbb{Z}) = 0$ (periodic cohomology [5, XII.7], [3, VI.9]), so (A5) gives $0 \to \mathbb{Z}/|G| \to \mathrm{rSH}^3(G) \to \mathrm{H}^2(G;\mathbb{Z}/2) \to 0$; and $\mathrm{H}^2(\mathbb{Z}/m;\mathbb{Z}/2)$ is $\mathbb{Z}/2$ or 0, $\mathrm{H}^2(Q_8;\mathbb{Z}/2) \cong (\mathbb{Z}/2)^2$, $\mathrm{H}^1(\mathbb{Z}/m;\mathbb{Z}/2) = 0$ for m odd. (2) The characters are $Z(1,1) = \sum_p q^{p^2/2}/\eta$ and $Z(1,g) = \sum_p \zeta^{pr} q^{p^2/2}/\eta$ with $\zeta = e^{2\pi i/m}$. Poisson summation gives $(Z(1,g)/Z(1,1))|S = \sum_p q^{(p+r/m)^2/2} / \sum_p q^{p^2/2}$, and T^L multiplies $q^{(p+r/m)^2/2}$ by $e^{2\pi i r^2/m'}$ and fixes the denominator. As e_m is a homomorphism, ν_m has order at least $m' = |\mathrm{rSH}^3(\mathbb{Z}/m)|$. □

2.6. **The orbifold.** For a non-anomalous G, [14, Thm. 4.2] constructs the orbifold from a trivialisation, and the proof of [14, Thm. 5.6] computes its Ramond–Ramond partition function as a sum over twisted sectors. For bosonic holomorphic vertex operator algebras the corresponding statements are theorems [17, 8, 20]; in the super case their ingredients exist ([20, Thms. 4.15, 4.17] for the $G \times \langle\sigma\rangle$-crossed structure, [4] for regularity), but their assembly is only sketched in [14]. We state what we use as a hypothesis.

Hypothesis 2.5 (ORB)**.** Let V be a strongly rational holomorphic vertex operator superalgebra and G a finite solvable group acting faithfully on V by even automorphisms that fix the conformal vector, with $G \cap \langle\sigma\rangle = 1$. If $\alpha(V,G) = 0$, then every trivialisation determines a strongly rational holomorphic vertex operator superalgebra $W = V/\!\!/G \supseteq V^G$ with the same conformal vector, such that, as $\mathbb{Z}/2$- and L_0-graded spaces,

$$W = \bigoplus_{[g]} V(g)^{C(g)}, \qquad W_R = \bigoplus_{[g]} V(g\sigma)^{C(g)},$$

the sums running over the conjugacy classes of G. Here, for each representative g, the centraliser $C(g)$ acts on $V(g)$ and on $V(g\sigma)$ by genuine representations ρ_g through x-intertwiners ($x \in C(g)$), and ρ_1 is the given action.

In Theorem 1.4 only the following parts are used: for g of even order, that the summand of W_R belonging to $[g]$ is a graded subspace of $V(g\sigma)$; for g of odd order, the formula $V(g\sigma)^{C(g)}$ with a genuine action. In Proposition 4.6 the NS decomposition is used for a hypothetical orbifold. In both places the parities of the summands play no role.

2.7. **Further standard inputs.**

(A8) Lifts of lattice automorphisms [18], [12, Ch. 10], [2, Prop. 4.1]: the automorphisms of V_Λ preserving $M(1)$ and inducing g on $\Lambda \otimes \mathbb{C}$ are the maps $u \otimes e^\lambda \mapsto \eta(\lambda)\, gu \otimes e^{g\lambda}$, and $\widehat{\Lambda}[2].O(\Lambda)$ consists of those with $\eta(\lambda) = \pm 1$.

(A9) The NS character of a strongly rational holomorphic vertex operator superalgebra is S-invariant: $\mathrm{ch}\, W(-1/\tau) = \mathrm{ch}\, W(\tau)$. Indeed it spans a one-dimensional space of trace functions stable under S [10, Thms. 4.12, 7.7], and $\mathrm{ch}\, W(i) > 0$.

(A10) Twisted modules of lattice vertex superalgebras [2, Thm. 4.2, Rem. 4.4]: existence, structure and characters (Proposition 2.3); on a g-twisted module $e^{2\pi i L_0}$ times the parity implements $g^{\pm 1}$; and modular covariance of twisted–twined traces for commuting pairs that generate a cyclic group [10, Thms. 4.12, 7.7].

3. Cyclic groups

This section is [14, Ex. 4.4] and the anomaly half of the proof of [14, Thm. 5.6], with two misprints corrected and one citation replaced by a proof (Remark 3.5).

Proposition 3.1. *For every $G \subseteq \mathrm{Aut}(C)$, the Majorana layer of $\alpha(V,G)$ is the restriction of the character ε. In particular it vanishes for every group of Table 1.*

Proof. The first claim is (A6) with Lemma 2.2. For the second, ε is a character, so generators suffice: $N_-(-1) = n$; $N_-(J) = N_-(K) = n/2 = 6k$ by (1); elements of odd order have $\varepsilon = 1$; and Jh, $-h$ are products of these. □

Proposition 3.2. *Let $g \in \mathrm{Aut}(C)$ have order m, no fixed vector, and $\varepsilon(g) = 1$. Then $\alpha(V,\langle g\rangle) = a\,\nu_m$ with $a/m' \equiv L\rho_g \pmod 1$.*

Proof. $Z(1,1) = \Theta_\Lambda/\eta^n$ is invariant under S and T^L (Λ is unimodular, L is even, and $Ln/24 = Lk/2 \in \mathbb{Z}$), so by (A7) $e_m(\alpha)$ is the eigenvalue of ST^LS^{-1} on $Z(1,g) = 1/\eta_g$ (Proposition 2.3). Since $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$ and $\sum_k d_k = 0$, $(1/\eta_g)|S = \prod_k k^{d_k/2} \prod_k \eta(\tau/k)^{-d_k}$. Every k with $d_k \neq 0$ divides m and hence L, so T^L multiplies this by $\exp(-2\pi i \frac{L}{24}\sum_k d_k/k) = e^{2\pi i L\rho_g} e^{-2\pi i Ln/24} = e^{2\pi i L\rho_g}$. Conclude with Lemma 2.4(2). □

Proposition 3.3. *Let $g \in \mathrm{Aut}(C)$ have odd order m. Then $\alpha(V,\langle g\rangle) = a\,\nu_m$ with $a/m \equiv 2m\rho_g \pmod 1$. If $h \in \mathrm{Aut}(C)$ has order 3 and no fixed coordinate, then $\alpha(V,\langle h\rangle) = 2k\,\nu_3$.*

Proof. By (A8), $s(g)$ acts on e^λ for $\lambda \in \Lambda^g$ by a sign $\eta(\lambda)$ with $\eta(\lambda)^m = 1$, so $\eta = 1$ on Λ^g ($s(g)$ is *standard* [18]), and $Z(1,g) = \Theta_{\Lambda^g}/\eta_g$. Poisson summation gives $\Theta_{\Lambda^g}(-1/\tau) = (-i\tau)^{r/2}\,\mathrm{vol}(\Lambda^g)^{-1}\Theta_{(\Lambda^g)^*}(\tau)$, $r = \mathrm{rk}\,\Lambda^g$. As $\Lambda^g = \Lambda \cap (\Lambda\otimes\mathbb{R})^g$ is primitive and Λ unimodular, $(\Lambda^g)^* = \pi_g(\Lambda)$ for the orthogonal projection $\pi_g = \frac{1}{m}\sum_j g^j$; and $m|\pi_g\lambda|^2 = \sum_j \langle\lambda, g^j\lambda\rangle \in \mathbb{Z}$, so T^{2m} fixes $\Theta_{(\Lambda^g)^*}$. The eta factor behaves as in Proposition 3.2, the powers of $-i\tau$ cancel, and $Z(1,1)$ is invariant under $ST^{2m}S^{-1}$. This gives $a/m \equiv 2m\rho_g$. For h the cycles have length 3 and $\sigma = 1$ (as $h^3 = 1$), so the Frame shape is $3^{n/3}$, $\rho_h = n/27$, and $2m\rho_h = 6n/27 = 8k/3$; hence $a \equiv 8k \equiv 2k \pmod 3$. The Majorana layer vanishes since m is odd. □

Corollary 3.4. *For the elements used in Table 1 the anomaly of the cyclic group they generate is $2k\,\nu_m$ in $\mathrm{rSH}^3(\mathbb{Z}/m) \cong \mathbb{Z}/m'$; it vanishes iff $m \mid k$:*

element	m	*signed cycles*	ρ_g	m'	α
-1	2	1-*cycles*, $\sigma=-1$	$3k/4$	4	$2k$
x *with* $x^2=-1$ (J,K,JK)	4	2-*cycles*, $\sigma=-1$	$9k/16$	8	$2k$
$-h$	6	3-*cycles*, $\sigma=-1$	$19k/36$	12	$2k$
Jh	12	6-*cycles*, $\sigma=-1$	$73k/144$	24	$2k$
h	3	3-*cycles*, $\sigma=1$	$4k/9$	3	$2k$

Proof. The cycle types follow from (1): $(-h)^3 = -1$, where 1-cycles are excluded because $(-h)^2 = h^2$ has no fixed coordinate; and $(Jh)^6 = -1$ with $(Jh)^4 = h$, which excludes 2-cycles of Jh in the same way. For the even types all cycles have length $\ell = m/2$ and $\sigma = -1$, so $\rho_g = \frac{n}{24}(1 + \frac{1}{2\ell^2}) = \frac{k}{2} + \frac{k}{m^2}$, and Proposition 3.2 gives $a \equiv m^2k + 2k \equiv 2k \pmod{2m}$, as mk is even. The last row is Proposition 3.3. □

Remark 3.5 (comparison with [14])**.** (1) For m odd the exponent in [14, Ex. 4.4] should be $\frac{2\pi i}{24}\sum_k d_k \frac{2m}{k}(k^2-1)$ (the operator is $ST^{2m}S^{-1}$); for $m = 3$ the conclusion of [14] is unaffected. (2) In the proof of [14, Thm. 5.6] each block contributes 6 times its number of sign changes to $\langle\chi, (1-g)\chi\rangle$ (Lemma 2.2), not $3(m/2) - 2\times$ odd; the conclusion $\langle\chi,(1-g)\chi\rangle \in 4\mathbb{Z}$ is unaffected.

(3) For $m = 3$, [14] cite [21, Ch. 5] for the Frame-shape formula of a standard lift; that source treats even lattices, while $\Lambda(C)$ is odd. Proposition 3.3 covers every unimodular lattice. (4) The extra hypothesis of [14, Ex. 4.4] that the lift has order m is automatic here, since all lifts of a g without fixed vector are conjugate to $s(g)$.

4. The quaternion group

Throughout this section C, J, K are as in Theorem 1.2, $n = 12k$, and $\alpha_C := \alpha(V_{\Lambda(C)}, Q_8) \in \widetilde{\mathrm{SH}}^3(Q_8)$. The six elements $\pm J, \pm K, \pm JK$ satisfy $x^2 = -1$ (note $(JK)^2 = -J^2K^2 = -1$), so none of them has a fixed coordinate or a fixed vector, and $Q_8/\langle -1\rangle$ acts freely on the coordinates, with $3k$ orbits of size 4.

4.1. **Restricted supercohomology of Q_8.** We use the following standard facts [5, XII.7], [1, IV], [3, VI.9].

Lemma 4.1. (1) $\mathrm{H}^2(Q_8; \mathbb{Z}/2) \cong (\mathbb{Z}/2)^2$, *and* $\mathrm{H}^3(Q_8; U(1)) \cong \mathrm{H}^4(Q_8; \mathbb{Z}) \cong \mathbb{Z}/8$, *generated by the second Chern class e of the 2-dimensional irreducible representation ρ_2.*

(2) *For $u \in \{i, j, k\}$ the restriction $\mathrm{H}^3(Q_8; U(1)) \to \mathrm{H}^3(\langle u\rangle; U(1)) \cong \mathbb{Z}/4$ is onto. Under the identifications $u \mapsto 1$, the three restrictions of e are equal, because e is invariant under the automorphism $i \mapsto j \mapsto k \mapsto i$ (ρ_2 is the only 2-dimensional irreducible representation).*

Identify each $\mathrm{rSH}^3(\langle u\rangle)$, $u \in \{i, j, k\}$, with $\mathbb{Z}/8$ through ν_4 and $u \mapsto 1$, and let $R = (\mathrm{res}_{\langle i\rangle}, \mathrm{res}_{\langle j\rangle}, \mathrm{res}_{\langle k\rangle})\colon \mathrm{rSH}^3(Q_8) \to (\mathbb{Z}/8)^3$.

Proposition 4.2. $\ker R = \{0, \kappa\}$ *with $\kappa := \iota_1(4e) \neq 0$. The class κ is fixed by every automorphism of Q_8.*

Proof. $|\mathrm{rSH}^3(Q_8)| = 32$ (Lemma 2.4). In $\mathrm{rSH}^3(\langle u\rangle) \cong \mathbb{Z}/8$ the image of ι_1 is the subgroup $2\mathbb{Z}/8$ of order 4, and by Lemma 4.1 $R(\iota_1 e) = (2c, 2c, 2c)$ for one odd c. For a character $\chi\colon Q_8 \to \{\pm 1\}$ let b_χ be the anomaly of Q_8 acting on $\mathrm{Fer}(2)$ through χ, i.e. by the rotation through π where $\chi = -1$. It lies in $\mathrm{rSH}^3(Q_8)$ by (A6), and by Lemma 2.4(2) its restriction to $\langle u\rangle$ is 0 if $\chi(u) = 1$ and $\nu_{4,2} = 4\nu_4$ if $\chi(u) = -1$. So $R(b_{\chi_i}) = (0, 4, 4)$, $R(b_{\chi_j}) = (4, 0, 4)$, $R(b_{\chi_k}) = (4, 4, 0)$, where χ_u has kernel $\langle u\rangle$. In $(2\mathbb{Z}/8)^3 \cong (\mathbb{Z}/4)^3$ the element $(1, 1, 1)$ generates a subgroup of order 4, $(0, 2, 2)$ is not in it, and $(2, 0, 2)$ is not of the form $(t, t + 2s, t + 2s)$; so these four elements generate a subgroup of order 16, and $|\ker R| \leq 2$. Finally ι_1 is injective, $4e \neq 0$, and $R(\kappa) = 4R(\iota_1 e) = 0$. The last claim holds because every automorphism of Q_8 fixes e. □

Lemma 4.3. $\alpha_C \in \mathrm{rSH}^3(Q_8)$; $\mathrm{res}_{\langle x\rangle}\alpha_C = 2k\,\nu_4$ *for $x \in \{\pm J, \pm K, \pm JK\}$, and* $\mathrm{res}_{\langle -1\rangle}\alpha_C = 2k\,\nu_2$. *Hence $\alpha_C \neq 0$ if $4 \nmid k$, and $\alpha_C \in \{0, \kappa\}$ if $4 \mid k$.*

Proof. Proposition 3.1, Corollary 3.4, (A2) and (A5), and Proposition 4.2. □

4.2. **Twisted sectors.** Let $\theta = s(-1)$; on each frame factor $V_{\sqrt{3}\mathbb{Z}}$ it is the reflection lift. We use the following q-series, all carrying $q^{L_0 - c/24}$:

$$e(\tau) = \frac{\eta(\tau)^n}{\eta(2\tau)^n}, \quad f(\tau) = \frac{\eta(2\tau)^{n/2}}{\eta(4\tau)^{n/2}}, \quad E(\tau) = 2^{n/2} q^{n/48} \prod_{j \text{ odd}} (1 - q^{j/2})^{-n},$$

$$g(\tau) = 2^{n/4} q^{n/192} \prod_{j \text{ odd}} (1 - q^{j/4})^{-n/2}, \quad F(\tau) = q^{n/48} \prod_{r \text{ odd}} (1 + q^r)^{-n/2}.$$

By Proposition 2.3, $e = Z(1, -1)$ and $f = Z(1, x)$ for every x with $x^2 = -1$ (Frame shapes $1^{-n}2^n$ and $2^{-n/2}4^{n/2}$).

Lemma 4.4. (1) *$V(-1) = S[\theta]\otimes T$, where $S[\theta]$ is the Fock space of the Heisenberg modes $\alpha(r)$, $r\in\frac12+\mathbb{Z}$, and T is the spinor module of the Clifford algebra $\mathrm{Cl}(n)$ generated by odd zero modes $\gamma_1,\dots,\gamma_n$, one for each frame factor; $\dim T = 2^{n/2}$, all of T has weight $n/16$, and $\operatorname{ch} V(-1) = E$. $V(-1)$ is irreducible already as a module over the frame.*

(2) *For $g\in Q_8$, every operator on $V(-1)$ implementing g is a multiple of $S[g]\otimes\phi_g$, where $S[g]$ is the action of g on the Fock space and $\phi_g\in\mathrm{Pin}(n)$ satisfies $\phi_g\gamma_i\phi_g^{-1} = s_i\gamma_{p(i)}$.*

(3) *There is a representation $\tilde\rho\colon Q_8\to GL(T)$ with $\tilde\rho(g)\in\mathbb{C}^\times\phi_g$ and $T\cong(\rho_2\oplus 2\cdot\mathbf{1})^{\otimes 3k}$. Hence $t_x := \operatorname{Tr}_T\tilde\rho(x) = 2^{3k}$ for $x\in\{\pm J,\pm K,\pm JK\}$ and $\operatorname{Tr}_T\tilde\rho(-1) = 0$. Every representation ρ with $\rho(g)\in\mathbb{C}^\times\phi_g$ is $\psi\otimes\tilde\rho$ for a character $\psi\colon Q_8\to\{\pm1\}$.*

(4) *For $x\in\{\pm J,\pm K,\pm JK\}$, $V(x) = S[x]\otimes T_x$ with T_x the spinor module of $\mathrm{Cl}(n/2)$, one zero mode for each signed 2-cycle of x; $\dim T_x = 2^{n/4}$, the superdimension of T_x is 0, the weights lie in $h_x+\frac14\mathbb{Z}_{\ge0}$ with $h_x = 3n/64$, and $\operatorname{ch} V(x) = g$.*

(5) *For $x\in\{\pm J,\pm K,\pm JK\}$ the trace of $S[x]$ on $S[\theta]$, weighted by $q^{L_0-c/24}$ with L_0 including the vacuum energy $n/16$ of $V(-1)$, is F.*

Proof. (1) On the frame, $\sqrt{3}\mathbb{Z}^n$ has a unique irreducible θ-twisted module, $S[\theta]\otimes T$ with $\dim T = |\sqrt{3}\mathbb{Z}^n/2\sqrt{3}\mathbb{Z}^n|^{1/2} = 2^{n/2}$, since n is even [2, Thm. 4.2, Rem. 4.4]; its zero modes are generated by the γ_i, which anticommute. θ-twisted frame modules are semisimple [2, Thm. 4.2], and $V(-1)$ has a ground-state space of dimension $|\det(1-(-1))|^{1/2} = 2^{n/2}$ (Proposition 2.3, $\varepsilon(-1)=1$), so as a frame module it is this irreducible module. The vacuum energy is $\rho_{-1} = n/16$. (2) An implementing operator conjugates the Heisenberg modes by g and the zero modes as stated; by Schur's lemma for the irreducible frame module it is unique up to a scalar. (3) On a Q_8-orbit O of coordinates, Q_8 acts on $\mathbb{R}^4_O$ by signed permutations, permuting $\pm e_p$ simply transitively; sending $x\in Q_8$ to xe_p for a fixed $p\in O$ identifies $\mathbb{R}^4_O$ with the quaternions, Q_8 acting by left multiplication. So Q_8 acts through $SU(2)_L\subset SO(4)$ (or $SU(2)_R$), which lifts isomorphically to $\mathrm{Spin}(4) = SU(2)_L\times SU(2)_R$; on the spinor module $S^+\oplus S^- = \mathbb{C}^2\oplus\mathbb{C}^2$ of $\mathrm{Cl}(4)$ the lift acts as $\rho_2\oplus 2\cdot\mathbf{1}$. The lifts are even Clifford elements, so they act factorwise on $T = \bigotimes_O S_O$ without Koszul signs, and their tensor product is a representation $\tilde\rho$. The character of ρ_2 is 0 on elements of order 4 and -2 at -1, so $t_x = 2^{3k}$ and $\operatorname{Tr}\tilde\rho(-1) = 0$. Two representations that agree up to scalars differ by a character of Q_8, which is ±1-valued. (4) As in (1), with $\sqrt{3}\mathbb{Z}^n/(1-x)\sqrt{3}\mathbb{Z}^n\cong(\mathbb{Z}/2)^{n/2}$ generated by one $\sqrt{3}e_i$ for each 2-cycle, and $|\det(1-x)| = 2^{n/2}$; the vacuum energy is $\rho_x = 3n/64$ and the Fock modes lie in $\pm\frac14+\mathbb{Z}$. (5) x has the eigenvalues $\pm i$ on $\Lambda\otimes\mathbb{C}$, each $n/2$ times, so the Fock trace is $\prod_{r\in\frac12+\mathbb{Z}_{\ge0}}(1+q^{2r})^{-n/2}$, and $q^{n/16-n/24} = q^{n/48}$. □

Lemma 4.5. *Write $(\phi|S)(\tau) = \phi(-1/\tau)$ and $(\phi|T)(\tau) = \phi(\tau+1)$. Then*

$$e|S = E,\qquad f|S = g,\qquad (g|T^2)|S = N\cdot F,\quad \text{where } N = \big(\sqrt{2}\,e^{2\pi i/48}\big)^{n/2} = 2^{3k}e^{2\pi ik/8}.$$

Equivalently $f(\tau/(1-2\tau)) = N\cdot F(\tau)$.

Proof. The first two follow from $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$; for example

$$f(-1/\tau) = 2^{n/4}\big(\eta(\tau/2)/\eta(\tau/4)\big)^{n/2} = g(\tau).$$

For the third, use $\eta(z+1) = e^{2\pi i/24}\eta(z)$ and Euler's identity

$$\eta\big(w+\tfrac12\big) = e^{2\pi i/48}\,\frac{\eta(2w)^3}{\eta(w)\,\eta(4w)},\qquad \text{i.e.}\qquad \prod_m\big(1-(-q)^m\big) = \frac{\prod_m(1-q^{2m})^3}{\prod_m(1-q^m)\prod_m(1-q^{4m})}.$$

With $w=\tau/4$ they give

$$\frac{\eta(\tau/2+1)}{\eta(\tau/4+\frac12)} = e^{2\pi i/48}\,\frac{\eta(\tau)\eta(\tau/4)}{\eta(\tau/2)^2},\qquad \text{so}\qquad g(\tau+2) = N\Big(\frac{\eta(\tau)\eta(\tau/4)}{\eta(\tau/2)^2}\Big)^{n/2} = N\,F(-1/\tau),$$

using $F = \big(\eta(\tau)\eta(4\tau)/\eta(2\tau)^2\big)^{n/2}$ and the S-law of η (the factors $\sqrt{-i\tau}$ cancel). Applying S gives $(g|T^2)|S = N\,F$, and $f|ST^2S = (g|T^2)|S$. □

4.3. **The character of a would-be orbifold.**

Proposition 4.6. *Assume (A1), (A9), (A10) and Hypothesis ORB. If $4 \mid k$ and the Q_8-action on $V_{\Lambda(C)}$ is non-anomalous, then $8 \mid k$.*

Proof. Let $W = V/\!\!/Q_8$ for some trivialisation. The conjugacy classes of Q_8 are $\{1\}$, $\{-1\}$, $\{\pm J\}$, $\{\pm K\}$, $\{\pm JK\}$, with $C(\pm 1) = Q_8$ and $C(x) = \langle x\rangle$ otherwise. We compute the contribution of each class to $\operatorname{ch} W$.

Untwisted sector. $\frac{1}{8}\sum_{g\in Q_8} Z(1,g) = \frac{1}{8}\big(Z(1,1) + e + 6f\big)$.

The class of -1. By Lemma 4.4(2),(3) and genuineness, $\rho_{-1}(g) = \psi(g)\,S[g] \otimes \tilde\rho(g)$ for a character ψ. The trace of $\rho_{-1}(g)$ is E for $g = 1$, 0 for $g = -1$, and $\psi(x)t_xF$ for $g = \pm x$ (note $\psi(-x) = \psi(x)$, since -1 is a commutator, and $t_{-x} = t_x$). So the contribution is $\frac{1}{8}E + \frac{1}{4}T_\psi F$ with $T_\psi = 2^{3k}\big(\psi(J) + \psi(K) + \psi(JK)\big)$.

The class of $x \in \{J, K, JK\}$. By (A10) and Schur's lemma, $\rho_x(x) = \epsilon\, e^{2\pi i L_0} P$ for a scalar ϵ, P the parity. On the weight space of weight $h_x + j/4$, which is (Fock level $j/4$) $\otimes T_x$, odd powers of $\rho_x(x)$ have trace 0 because T_x has superdimension 0, and $\rho_x(x)^2$ is the scalar $\epsilon^2 e^{4\pi i h_x}(-1)^j$. So the projector onto the invariants has trace $\frac{1}{4}(1 + \epsilon^2 e^{4\pi i h_x}(-1)^j)$ times the dimension, which is nonzero for $j = 0$ or $j = 1$; both levels are nonzero. Hence $W_x := V(x)^{\langle x\rangle} \neq 0$. On $W_x \subseteq W$ the operator $e^{2\pi i L_0}$ is $\pm P$ (W is a vertex operator superalgebra) and $\rho_x(x) = 1$, so $\epsilon = \pm 1$ and $\rho_x(x^2) = e^{4\pi i L_0}$, whose trace is $e^{4\pi i c/24} g(\tau+2) = g(\tau+2)$. The contribution is $\frac{1}{4}\big(g(\tau) + g(\tau+2)\big)$.

S-invariance. Altogether

$$\operatorname{ch} W = \tfrac{1}{8}Z(1,1) + \tfrac{1}{8}e + \tfrac{3}{4}f + \tfrac{1}{8}E + \tfrac{1}{4}T_\psi F + \tfrac{3}{4}\big(g + g|T^2\big).$$

$Z(1,1) = \Theta_\Lambda/\eta^n$ is S-invariant, and by Lemma 4.5 $e \leftrightarrow E$, $f \leftrightarrow g$, $g|T^2 \mapsto NF$ and $F \mapsto (g|T^2)/N$ under S. Hence

$$\operatorname{ch} W|S - \operatorname{ch} W = \tfrac{1}{4}\big(3N - T_\psi\big)\big(F - (g|T^2)/N\big).$$

By (A9) the left side vanishes. The leading exponents of F and $g|T^2$ are $n/48$ and $n/192$, so $F \neq (g|T^2)/N$ and $T_\psi = 3N$, i.e. $\psi(J) + \psi(K) + \psi(JK) = 3e^{2\pi i k/8}$. The left side is 3 ($\psi = 1$) or -1, so $\psi = 1$ and $e^{2\pi i k/8} = 1$. □

4.4. **Proof of Theorem 1.2.** If $4 \nmid k$, $\operatorname{res}_{\langle J\rangle} \alpha_C = 2k\nu_4 \neq 0$ (Lemma 4.3). If $k \equiv 4 \pmod 8$, then $\alpha_C \in \{0, \kappa\}$ by Lemma 4.3, and $\alpha_C = 0$ is excluded by Proposition 4.6; so $\alpha_C = \kappa \neq 0$. Let $8 \mid k$. Let D be a self-dual code of length 48 with $J_D, K_D \in \operatorname{Aut}(D)$ satisfying the same relations; for instance the sum of twelve Q_8-invariant tetracodes, Q_8 acting on each 4-set by left multiplication on quaternions [16, Lemma 6.2]. Then Q_8 acts on $C \oplus D$ through (J, J_D) and (K, K_D), and $s_{C\oplus D} = s_C \otimes s_D$, so $\alpha_{C\oplus D} = \alpha_C + \alpha_D$ by (A4). Both $C \oplus D$ and D have $k \equiv 4 \pmod 8$, so both anomalies equal κ, which does not depend on the identification of the groups with Q_8 (Proposition 4.2). Hence $\alpha_C = 0$. □

The proof shows more: $\alpha_C = (k/4)\,\kappa$ when $4 \mid k$.

Remark 4.7 (free fermions)**.** For a Q_8-invariant sum D_k of tetracodes of length $12k$ the lattice is $\mathbb{Z}^{12k}$ (the only odd unimodular lattice of rank 4 is $\mathbb{Z}^4$), so $V_{\Lambda(D_k)} = \operatorname{Fer}(8)^{\otimes 3k}$. The weight-$\frac{1}{2}$ space of each factor is spanned by the $e^{t/\sqrt{3}}$, t a nonzero tetracode word, which Q_8 permutes up to sign and freely; so it is the real regular representation $\operatorname{Reg} = \mathbf{1} \oplus \chi_i \oplus \chi_j \oplus \chi_k \oplus \mathbb{H}$. Let $a = \alpha(\operatorname{Fer}(\operatorname{Reg}))$. Then $\alpha_{D_k} = 3k\,a$, so $12a = \kappa$ and $24a = 0$ by the proof above; as $|\mathrm{rSH}^3(Q_8)| = 32$, a has order exactly 8 and $4a = \kappa$. Moreover $\alpha_C = 3k\,a$ for every C: the

difference $\delta(C) = \alpha_C - 3k\,a$ lies in $\ker R$ (Reg restricted to $\langle i\rangle$ rotates two planes through $\pi/2$ and one through π, so its anomaly there is $(1+1+4)\nu_4$, and $3k\cdot 6 \equiv 2k \pmod 8$), it is additive, it vanishes on the D_k, and it vanishes for $4 \mid k$; adding a D_j with $j \equiv -k \pmod 4$ gives $\delta(C) = 0$ in general.

In spin-bordism terms the anomaly of fermions in a real representation ρ on a spherical space form S^3/Γ with spin structure $\epsilon\colon \Gamma \to \{\pm 1\}$ is $\frac{1}{2}(\eta(\rho) - \dim \rho\,\eta(\mathbf{1})) \bmod 1$, with Donnelly's $\eta(\rho) = -\frac{1}{|\Gamma|}\sum_{g\neq 1}\epsilon(g)\chi_\rho(g)/(2-\operatorname{tr} g)$ [11]. For $\rho = \mathrm{Reg}$ this is $\frac{5}{8}$ on S^3/Q_8 for all four spin structures, $\frac{1}{4}$ on $S^3/\langle i\rangle$ and $\frac{1}{2}$ on $\mathbb{RP}^3$; one Majorana fermion odd under $\mathbb{Z}/2$ gives $\pm\frac{1}{8}$ on $\mathbb{RP}^3$. This is consistent with a of order 8 and with $4a$ vanishing on all cyclic subgroups. We do not use these values.

The twisted sector of Lemma 4.4 also gives a direct invariant. Normalise the lifts of J, K, JK on $V(-1)$ by transporting their twining characters on V along ST^2S (Lemma 4.5); for $4 \mid k$ these are the canonical lifts of the non-anomalous cyclic subgroups. They differ from $\tilde\rho$ by scalars, and $\hat J\hat K = \mu\,\widehat{JK}$ for a scalar μ; with the conventions of the computations in [23] one finds $\mu = e^{-2\pi i k/8} = \exp(2\pi i\cdot 3k\cdot\frac{5}{8})$, the value of $3k\,a$ on S^3/Q_8. Finally, $\alpha_C = 3k\,a$ can also be derived by gauging: a sum C_0 of Q_8-invariant tetracodes is a multi-neighbour of C, and $V_{\Lambda(C_0)} = V_{\Lambda(C)}/\!\!/\widehat{W}$ for a 3-group $\widehat{W}$ normalised by Q_8, so a coprime-gauging lemma [22, 25, 24] transports the anomaly from C to C_0. That route needs the super version of the gauging lemma, which the proof above avoids.

Remark 4.8. For $k \equiv 4 \pmod 8$ a non-anomalous Q_8 would produce an orbifold of index 3, which is not divisible by $24/\gcd(k,24) \in \{6,2\}$. So Theorem 1.2 is what the Stolz–Teichner conjecture predicts.

5. The group $Q_8 \times \mathbb{Z}/3$

Lemma 5.1 ([3, III.9–III.10])**.** *Let G be finite, M a trivial G-module, $q \geq 1$, and P a Sylow p-subgroup. Then res^G_P is injective on the p-primary part of $\mathrm{H}^q(G;M)$, and on all of $\mathrm{H}^q(G;\mathbb{Z}/2)$ if $p = 2$. Hence $\beta \in \mathrm{H}^q(G;M)$ vanishes iff all its Sylow restrictions vanish.*

Proposition 5.2. *Let G be finite and $\alpha \in \widetilde{\mathrm{SH}}^3(G)$. Then $\alpha = 0$ iff $\operatorname{res}^G_P \alpha = 0$ for every Sylow subgroup P.*

Proof. Suppose all Sylow restrictions vanish, and let P_2 be a Sylow 2-subgroup. By naturality $\operatorname{res}_{P_2}\operatorname{maj}(\alpha) = \operatorname{maj}(\operatorname{res}_{P_2}\alpha) = 0$, so $\operatorname{maj}(\alpha) = 0$ by Lemma 5.1, and $\alpha = \iota_2(\alpha')$ with $\alpha' \in \mathrm{rSH}^3(G)$; since ι_2 is injective for every subgroup, all Sylow restrictions of α' vanish. In the same way $\operatorname{gw}(\alpha') = 0$, so $\alpha' = \iota_1(\beta)$ with $\beta \in \mathrm{H}^3(G;U(1))$ whose Sylow restrictions vanish, and $\beta = 0$ by Lemma 5.1. □

Proof of Theorem 1.3. The Sylow subgroups of $G = Q_8 \times \langle h\rangle$ are Q_8 and $\langle h\rangle$, both normal. The restriction to Q_8 is α_C, which vanishes iff $8 \mid k$ (Theorem 1.2). An element of order 3 without fixed coordinate consists of 3-cycles, so the restriction to $\langle h\rangle$ is $2k\,\nu_3$ (Proposition 3.3), which vanishes iff $3 \mid k$. Conclude with Proposition 5.2. □

The same argument shows that $(\operatorname{res}_{Q_8}, \operatorname{res}_{\mathbb{Z}/3})\colon \widetilde{\mathrm{SH}}^3(Q_8\times\mathbb{Z}/3) \to \widetilde{\mathrm{SH}}^3(Q_8)\oplus\widetilde{\mathrm{SH}}^3(\mathbb{Z}/3)$ is an isomorphism (inflations give the inverse): there are no mixed terms.

6. The orbifold index

Throughout, C and G are as in Theorem 1.4, $V = V_{\Lambda(C)}$, G acts non-anomalously, and $W = V/\!\!/G$ is an orbifold as in Hypothesis ORB. Since the anomaly vanishes, so does its Majorana layer, hence $\varepsilon = 1$ on G (Proposition 3.1).

Lemma 6.1. (1) *Every commuting pair in G generates a cyclic subgroup.*
(2) *If $g \in G$ has even order m, then $g^{m/2} = -1$.*
(3) *Every $g \in G \setminus \{\pm 1\}$ has no fixed coordinate; so $\mathrm{tr}_{\mathbb{R}^n} g = 0$.*

Proof. (1) is (i). (2) $\langle g \rangle$ contains exactly one involution, $g^{m/2}$, and it is -1 by (ii). (3) A coordinate fixed by g is fixed by every power of g. By (iii) the order of g is $2^a 3^b$ with $b \le 1$. If $b = 1$, then g^{2^a} is a nontrivial element of $\langle h \rangle$, which has no fixed coordinate. If $b = 0$, then $a \ge 2$, since $g \ne \pm 1$ and -1 is the only involution; so $g^{2^{a-1}} = -1$ by (2), and a coordinate fixed by g, with $ge_i = \pm e_i$, would give $-e_i = g^{2^{a-1}} e_i = e_i$. A signed permutation matrix without fixed coordinate has trace 0. □

Lemma 6.2. *Let $g \in G$ have even order m. Then the eigenvalues of g on $\mathbb{R}^n$ are $e^{2\pi i s/m}$ with s odd, each with multiplicity $2n/m$; g has no fixed vector; and the lowest weight of $V(g\sigma)$ is*

$$E_g = \rho_g = \frac{n}{24} + \frac{n}{12m^2} > \frac{c}{24}.$$

Proof. By Lemma 6.1 the character of $\langle g \rangle$ on $\mathbb{R}^n$ is n at 1, $-n$ at $g^{m/2} = -1$, and 0 elsewhere, so $e^{2\pi i s/m}$ has multiplicity $\frac{n}{m}(1 - (-1)^s)$. With $m = 2p$, $\sum_{s \text{ odd} < m} \frac{s}{m}(1 - \frac{s}{m}) = \frac{2p^2+1}{12p}$, so $\rho_g = \frac{n}{4} \cdot \frac{2}{m} \cdot \frac{2p^2+1}{12p} = \frac{n}{24} + \frac{n}{12m^2}$. By Proposition 2.3 ($\varepsilon(g) = 1$) the lowest weight of $V(g\sigma) \cong V(g)$ is ρ_g. □

Lemma 6.3. *Every $V(g)$ and $V(g\sigma)$, $g \in G$, is of $\mathbb{C}$-type. If g has odd order and $x \in C(g)$, every x-intertwiner on $V(g\sigma)$ is even.*

Proof. The Majorana layer, which records which $V(g)$ are of Clifford type, is $\varepsilon = 1$; and V_R is of $\mathbb{C}$-type, with $V(g\sigma) \cong V(g) \boxtimes V_R$ in the $G \times \langle \sigma \rangle$-crossed supercategory of twisted modules [20, Thms. 4.15, 4.17], in which each component has one simple object, invertible. For a $\mathbb{C}$-type simple object M let $p(M, x) \in \mathbb{Z}/2$ be the parity of an x-intertwiner; it is well defined, $p(V, x) = 0$, and p is additive under $\boxtimes$ (an isomorphism $M \boxtimes N \cong P$ of either parity enters twice). So $a \mapsto p(V(g^a), x)$ is a homomorphism from $\mathbb{Z}/\operatorname{ord}(g)$ to $\mathbb{Z}/2$, trivial for odd order, and $p(V(g\sigma), x) = p(V_R, x)$. Finally $p(V_R, x) = 0$: an x-intertwiner maps e^{ν_w} to a multiple of $e^{\nu_{xw}}$, whose parity is $f(xw) = \varepsilon(x) f(w) = f(w)$ (if C has no maximal codeword, $\mathrm{Index}(C) = 0$ and there is nothing to prove). □

Lemma 6.4. *Let A be an x-intertwiner on V_R, $x \in G$, and $Z_{RR}(1, A) = \mathrm{str}_{V_R} A q^{L_0 - c/24}$.*
(1) *$Z_{RR}(1, A) = 0$ if x has even order.*
(2) *If x has order 3 and $A^3 = 1$, then $Z_{RR}(1, A) = \zeta\, T_x(C)$ for some $\zeta^3 = 1$.*

Proof. By Proposition 2.1(5) only the ground states e^{ν_w} contribute, and A maps $\mathbb{C}e^{\nu_w}$ onto $\mathbb{C}e^{\nu_{xw}}$, since it conjugates the Heisenberg zero modes by x. (1) A maximal codeword fixed by x would be fixed by $x^{m/2} = -1$, but $-w \ne w$. (2) The lift $s(x)$ acts on e^α, $\alpha \in \Lambda^x$, by a sign whose cube is 1, hence trivially. Any two x-fixed ground states $e^{\nu_w}, e^{\nu_{w'}}$ have momenta differing by some $\alpha \in \Lambda^x$, and the leading term of $Y_R(e^\alpha, z) e^{\nu_w}$ is a nonzero multiple of $e^{\nu_w + \alpha}$; since $A Y_R(e^\alpha, z) A^{-1} = Y_R(e^\alpha, z)$, A has one eigenvalue ζ on all x-fixed ground states, and $\zeta^3 = 1$. The parities are $f(w)$, so $Z_{RR}(1, A) = \zeta \sum_{xw=w} f(w) = \zeta\, T_x(C)$. □

Proof of Theorem 1.4. W contains $V^G \ni \tau$ with the same conformal vector, so it is $\mathcal{N} = 1$ with the supercurrent τ, and $G_0^2 = L_0 - c/24$ on W_R. Hence $\mathrm{Index}(W) = \mathrm{str}_{W_R} q^{L_0 - c/24}$ is the superdimension of the weight-$c/24$ part of W_R, which is the sum of the contributions of the classes $[g]$.

g of even order. The summand lies in $V(g\sigma)$, which has no states of weight $c/24$ by Lemma 6.2; it contributes 0.

g of odd order. The summand is $V(g\sigma)^{C(g)}$, the image of the projector

$$e_g = \frac{1}{|C(g)|}\sum_{x\in C(g)} \rho_g(x),$$

which is even by Lemma 6.3. Each $\rho_g(x)$ commutes with G_0, since x fixes τ, so the contribution is $\frac{1}{|C(g)|}\sum_{x\in C(g)} Z_{RR}(g,x)$ with $Z_{RR}(g,x) = \operatorname{str}_{V(g\sigma)} \rho_g(x) q^{L_0-c/24}$.

For $g=1$ this is $\frac{1}{|G|}\sum_x Z_{RR}(1,x) = \operatorname{Index}(C)/|G|$, by Lemma 6.4: elements of even order contribute 0, and the elements of order 3 are $h^{\pm1}$, with $T_{h^{\pm1}} = T_h = 0$.

Let $g \in \langle h\rangle \setminus \{1\}$; then $C(g) = G$ by (iii). If $x \in G$ has even order, then $\langle g, x\rangle$ is cyclic, and by modular covariance (A10) $Z_{RR}(g,x)$ is a multiple of the S-transform of $Z_{RR}(x, g^{-1})$, which vanishes because $V(x\sigma)$ has no states of weight $c/24$. If x has odd order, then $x \in \langle g\rangle$; as $SL_2(\mathbb{Z}) \to SL_2(\mathbb{Z}/3)$ is onto and $SL_2(\mathbb{Z}/3)$ acts transitively on $(\mathbb{Z}/3)^2 \setminus 0$, some $\gamma \in SL_2(\mathbb{Z})$ carries the pair (g,x) to $(1,y)$ with $y \neq 1$ in $\langle h\rangle$, and $Z_{RR}(1,y) = \zeta T_y(C) = 0$. So all twisted classes contribute 0, and $\operatorname{Index}(W) = \pm\operatorname{Index}(C)/|G|$, the sign being the convention for the parity of W_R. □

Remark 6.5. For $G = Q_8$, and for cyclic G of order 1, 2 or 4, every class except [1] has even order, so no modular transformation of twisted–twined traces is used. Changing the trivialisation can change individual twisted–twined terms by phases, but every term except $Z_{RR}(1,1)$ vanishes for every normalisation, so the index is the same for all orbifolds.

7. Proof of the Main Theorem, and remarks

Proof of Theorem 1.1. Let $m = \gcd(k, 24)$. By [16, Thms. B, C] there is a self-dual code C of length $12k$ with $\operatorname{Index}(C) = 24$ and the automorphisms of Table 1: for $m \in \{1,2\}$ any code of [16, Thm. B]; for $3 \mid k$ an h of order 3 without fixed coordinate and with $T_h(C) = 0$ [16, Thm. C(a)]; for $4 \mid k$ elements J, K with $J^2 = K^2 = -1$, $KJ = -JK$ [16, Thm. C(b)]; and for $12 \mid k$ all three, commuting [16, Thm. C(c)]. For $m \in \{1,2,3,4,6,12\}$ the group G is cyclic, generated by 1, -1, h, J, $-h$, Jh, of order m; by Corollary 3.4 its anomaly is $2k\nu_m = 0$, as $m \mid k$. For $m = 8$, $G = Q_8$ is non-anomalous by Theorem 1.2, and for $m = 24$, $G = Q_8 \times \langle h\rangle$ by Theorem 1.3. Each G satisfies (i)–(iii) of Theorem 1.4: its abelian subgroups are cyclic (for $Q_8 \times \mathbb{Z}/3$ they are products of subgroups of coprime orders); $-1 = g^{m/2}$ resp. the centre of Q_8 is its only involution; and its elements of odd order form 1 or $\langle h\rangle$, central, with $T_h(C) = 0$ ($T_{h^2} = T_h$, and $(Jh)^4 = h$). By Hypothesis ORB the orbifold $W = V_{\Lambda(C)}/\!\!/G$ exists, and by Theorem 1.4 it is a holomorphic $\mathcal{N}=1$ vertex operator superalgebra of central charge $12k$ and index $24/|G| = 24/m$. □

Remark 7.1 (reduction to $k \le 24$)**.** If W and W' are holomorphic $\mathcal{N}=1$ vertex operator superalgebras of central charges in $12\mathbb{Z}$, then so is $W \otimes W'$, with supercurrent $\tau \otimes \mathbf{1} + \mathbf{1} \otimes \tau'$; its Ramond sector is $W_R \otimes W'_R$, so $\operatorname{Index}(W \otimes W') = \pm\operatorname{Index}(W)\operatorname{Index}(W')$. With W_{24} the orbifold of index 1 at $k = 24$ and W_r that of index $24/\gcd(r,24)$ at $k = r \le 24$, the theory $W_{24}^{\otimes q} \otimes W_r$ has central charge $12(24q + r)$ and index $24/\gcd(24q + r, 24)$. So the Main Question needs only the codes of length at most 288; the chains of [16] are used for finitely many steps.

Remark 7.2 ($k = 24$ and TMF)**.** Let W be the orbifold of index 1 and central charge 288. Under the Stolz–Teichner conjecture [14, Conj. 5.4] it defines a class $[W] \in \pi_{576}\mathrm{TMF}$ mapping to $\pm\Delta^{24}$ in MF_{288}, so $[W] \mp \Delta^{24}$ lies in the kernel of $\pi_*\mathrm{TMF} \to \mathrm{MF}$, which is the torsion. Torsion elements of $\pi_*\mathrm{TMF}$ are nilpotent by the nilpotence theorem [6], their images in $\mathrm{MU}_*\mathrm{TMF}$ being torsion in a torsion-free group. Hence $[W] = \pm\Delta^{24}(1+t)$ with t nilpotent, a unit of $\pi_*\mathrm{TMF}$; and a theory of central charge 288 can define a unit only if its index is ±1. So at $k = 24$ the

Main Question asks for a holomorphic $\mathcal{N}=1$ theory representing the periodicity class Δ^{24} up to torsion, and Theorem 1.1 provides one.

Remark 7.3 (why the seed for $12 \mid k$ has length 72)**.** The codes for $12 \mid k$ need $Q_8 \times \mathbb{Z}/3$ and are built from a seed of length 72 [16, §7.3]. Q_8-invariant codes are $\mathbb{F}_3^2 \otimes U$ with U Lagrangian in $\mathbb{F}_3^{2r}$ for the symplectic form $\sum_b (u_{2b}v_{2b+1} - u_{2b+1}v_{2b})$ (cf. [16, App. A.2]), which makes their index computable at length 72. At length 36, in all our samples, $Q_8 \times \mathbb{Z}/3$-invariant codes with $T_h = 0$ have index divisible by 72, which rules out seeds of that length for chains with index 24; at length 72 this pattern fails, and a seed exists.

Remark 7.4 (formalisation)**.** The code-level content of Theorem 1.1, i.e. for every k a self-dual code of length $12k$ and index 24 with the group data of Table 1, is formalised in Lean 4 with Mathlib in the repository of [16], with the value 24 of the length-72 seed sum as an explicit hypothesis. The vertex-algebra side is not formalised.

Appendix A. Computations

The explicit identities of this paper were checked by computer, with the code data parsed from the source of [16]; the script and its log are in [23].

- The Ramond ground states of Proposition 2.1(4) (the per-coordinate minima, and the signed count 24 on the Golay code), Lemma 2.2 and (1) on random signed permutations, and the two expressions for ρ_g.
- The calibration of Lemma 2.4 (theta quotients under ST^LS^{-1}, numerically); the eigenvalues of Propositions 3.2 and 3.3 numerically (eta quotients for $n = 12, 24$; the theta series of Λ^h for an element h of order 3 without fixed coordinate on the Golay code and on E_{36}); and the anomalies $2k\nu_m$ of Corollary 3.4 exactly, for $k \le 48$, and for every element of every group of Table 1 on the codes of [16], together with $\varepsilon = 1$ and the Frame shapes.
- The cohomology of Q_8 in low degree over $\mathbb{F}_2$, and the order 16 of the subgroup of Proposition 4.2.
- Lemma 4.4(3): explicit Pin lifts in Cl(4) on every Q_8-orbit of coordinates of the Golay code and of E_{72} (even, unique up to scalars, with exactly four linearisations $\rho_2 \oplus 2\psi$), and on the whole spinor module of Cl(12) for the Golay code (traces $8\psi(x)$ at elements of order 4 and 0 at -1).
- The three identities of Lemma 4.5 (numerically at several τ, for $n = 12, 24, 48$); the formula for $\operatorname{ch} W|S - \operatorname{ch} W$ in Proposition 4.6 numerically for $k = 4, 8$ and all four ψ, and its zero set ($8 \mid k$, $\psi = 1$) exactly.
- The eta invariants of Remark 4.7, in exact rational arithmetic.
- The eigenvalue multiplicities and the vacuum energies E_g of Lemma 6.2 for every element of even order, and the group theory of Lemma 6.1, for every group of Table 1 on the codes of [16]; and the arithmetic of Table 1 and Remark 7.1.

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